

Lucas Castillo



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Education

- 2021 – (awaiting viva) **Ph.D. in Psychology** UNIVERSITY OF WARWICK, UK.
Supervisor: Prof. Adam Sanborn
Thesis title: *A Local Sampling Account of Random Generation*
- Jul-Sep 2024 **Research visit** at UNIVERSITY OF HAMBURG, Germany
under the supervision of Prof. Sebastian Gluth.
- 2019 – 2020 **M.Sc. Cognitive and Decision Sciences** UNIVERSITY COLLEGE LONDON, UK
Dissertation title: *Does newly learned causal knowledge guide the perception of temporal order?*
- 2012 – 2016 **B.Sc. Psychology** UNIVERSIDAD DE SEVILLA, Spain.

Acquired Funding

- 2025 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Rumination and Random Generation*
- 2024 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Exploration Behaviour in Decisions from Experience*

Publications

Journal Articles

- 2 Sanborn, A. N., Zhu, J.-Q., Spicer, J., León-Villagrà, P., **Castillo, L.**, Falbén, J. K., Li, Y.-X., Tee, A., & Chater, N. (2025). Noise in Cognition: Bug or Feature? *Perspectives on Psychological Science*, 20(3), 572–589. <https://doi.org/10.1177/17456916241258951>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). Explaining the flaws in human random generation as local sampling with momentum. *PLOS Computational Biology*, 20(1), 1–24. <https://doi.org/10.1371/journal.pcbi.1011739>

Preprints

- 6 Castillo, L., León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025a). *People Need a Few Seconds to Be Random*. PsyArXiv. <https://doi.org/10.31234/osf.io/awg9j>.
- 5 **Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025b). *Random Generation is What Comes to Mind*. PsyArXiv. <https://doi.org/10.31234/osf.io/c9xq5>. **Under review at Cognition.**
- 4 **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2025). *The samplr package: A tool for modeling human cognition with sampling algorithms*. PsyArXiv. <https://doi.org/10.31234/osf.io/ax8hm>. **Submitted to Behavior Research Methods.**

Publications (continued)

- 3 Li, Y.-X., Falben, J., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Qian, S., Chater, N., & Sanborn, A. N. (2025). *Does utility bias mental simulations of risky events?* PsyArXiv. <https://doi.org/10.31234/osf.io/awdr7>. **Under review at JEP: General.**
- 2 Falbén, J., **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). *Biased mind or biased world? Assessing the accuracy of cultural beliefs that underlie social judgments.* PsyArXiv. <https://doi.org/10.31234/osf.io/5hdbp>. **Under review at Memory & Cognition.**
- 1 León-Villagrà, P., **Castillo, L.**, Falbén, J., Chater, N., & Sanborn, A. N. (2024). *Eliciting Beliefs with Random Generation Tasks.* <https://doi.org/10.31234/osf.io/687au>

Peer-reviewed Conference Proceedings Articles

- 4 Valentin, S.[†], **Castillo, L.**[†], Sanborn, A. N., & Lucas, C. G. (2024). Distinguishing Between Process Models of Causal Learning. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2p43j2cw>
- 3 Li, Y.-X., Falbén, J. K., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Chater, N., & Sanborn, A. N. (2024). Probability, but not utility, influences repeated mental simulations of lotteries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2ps2t4sz>
- 2 León-Villagrà, P., **Castillo, L.**, Chater, N., & Sanborn, A. N. (2022). Eliciting Human Beliefs using Random Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. <https://escholarship.org/uc/item/88k526dj>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2021). Local Sampling with Momentum Accounts for Human Random Sequence Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. <https://escholarship.org/uc/item/3gz154kg>

† = equal contribution

Conferences

Talks

- July 2024 **MathPsych** Characterizing People's Sampling Engines Using Random Generation
- August 2023 **SPUDM** Production rate effects on random generation elicitation
- July 2023 **MathPsych** Modelling speeded random generation as sampling for inference.
- July 2022 **Virtual MathPsych** Using sampling algorithms to explain human random generation.
- October 2021 **Computational Cognition Workshop** Human random generation as a locally-bound process.

Poster Presentations

- July 2024 **Annual Meeting of the Cognitive Science Society** People Need About Five Seconds to be Random: Autocorrelated Sampling Algorithms Can Explain Why
- November 2021 **Annual Meeting of the Psychonomic Society** Local sampling can explain human deviations from randomness.
- July 2021 **Annual Meeting of the Cognitive Science Society** Local sampling with momentum accounts for human random sequence generation.

Reviews

Since	Venue	Count
2024	Journal of Experimental Psychology: General	1
2023	Journal of Experimental Psychology: Learning, Memory, and Cognition	1
2024	Cogsci Conference	10
2023	SPUDM Conference	5

Misc.

Other Outputs

- R Packages **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplr*: Compare Human Performance to Sampling Algorithms. <https://CRAN.R-project.org/package=samplr>.
- Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplrData*: Datasets from the SAMPLING Project. <https://CRAN.R-project.org/package=samplrData>.

Other University Involvement

- 2024 – present **Member of ReproducibiliTea Organizing Committee** at UNIVERSITY OF WARWICK, UK (University-wide Open Research initiative)
- 2022 – 2023 **Student Representative** for Psychology PhD Students at UNIVERSITY OF WARWICK, UK
- 2021 – 2023 **Teaching Assistant**, UNIVERSITY OF WARWICK, UK.
Modules: *Methods and Analysis in Behavioural Science* and
Bayesian Approaches in Behavioural Science
- 2019 – 2020 **Student Representative** for M.Sc. Cognitive and Decision Sciences cohort at UNIVERSITY COLLEGE LONDON, UK

Awards and Scholarships

- 2025 **Best PhD Student Paper** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK.
Granted for Castillo et al. 2024.
- 2021 **Departmental PhD Scholarship** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK
- 2020 **Best performing student** of the academic year for the MSc in Cognitive and Decision Sciences, UNIVERSITY COLLEGE LONDON, UK.

Skills

- Languages Spanish (mother tongue), Catalan, English (fluent), French, German (learning!)
- Coding git, Python, R, C++, L^AT_EX, HTML, css, JavaScript, REACTJS.

References

Available upon request